

Task 2 - Exploratory Data Analysis (EDA) Project Report

This project focuses on performing Exploratory Data Analysis (EDA) on an automobile dataset using Python. The goal of the project was to clean the dataset, identify patterns, detect trends, analyze relationships between variables, and derive meaningful business insights using statistical and visual analysis techniques.

Project Objectives

- Understand the structure and quality of the automobile dataset.
- Perform data cleaning and preprocessing.
- Handle missing values and duplicate records.
- Analyze univariate, bivariate, and multivariate relationships.
- Identify correlations affecting automobile prices.
- Detect market trends and outliers.
- Generate actionable insights using data visualization.

Dataset Overview

The dataset contains multiple automobile-related features such as price, engine size, fuel type, body style, horsepower, drive wheels, bore ratio, curb weight, and other vehicle specifications. The dataset was imported using pandas and explored using methods such as: **info()** for data types and null value inspection **describe()** for statistical summaries **duplicated()** for duplicate detection **isnull()** for missing value analysis

Data Cleaning & Preprocessing

Several preprocessing operations were performed to improve data quality: Checked dataset structure and feature data types. Detected and handled missing values. Converted object columns into numerical formats where required. Filled missing values in categorical columns using mode. Validated data consistency after preprocessing. Removed or investigated duplicate records.

Exploratory Data Analysis

The project performed three major forms of analysis:

- **Univariate Analysis:** Distribution analysis of price, brand frequency, and body styles using histograms and count plots.
- **Bivariate Analysis:** Relationship analysis between engine size and price, body style and price, and correlations between numerical variables.
- **Multivariate Analysis:** Combined analysis using heatmaps, regression plots, grouped comparisons, and segmented market analysis.

Key Insights Generated

- The automobile market is highly right-skewed, meaning most vehicles fall in the affordable range while luxury vehicles form a small premium segment.
- Toyota dominated the dataset, indicating strong market representation and customer preference.
- Sedans and hatchbacks were the most common body styles in the dataset.
- Engine size showed a very strong positive relationship with price, proving it to be one of the most influential pricing factors.
- Rear-wheel-drive (RWD) vehicles generally occupied the premium market segment.
- Luxury and high-performance vehicles appeared as statistical outliers due to exceptionally high prices.
- Fuel type, curb weight, and horsepower significantly affected overall vehicle valuation.

Major Achievements in This Project

- **Successful End-to-End EDA Pipeline:** Completed the entire data analysis lifecycle from raw dataset import to final business insights.
- **Efficient Data Cleaning:** Managed missing values, inconsistent data types, and dataset validation successfully.
- **Advanced Visualization:** Implemented histograms, KDE plots, regression plots, scatter plots, heatmaps, and boxplots for deep insight generation.
- **Correlation Discovery:** Identified strong relationships between engine size, horsepower, curb weight, and automobile pricing.
- **Outlier Detection:** Successfully identified premium/luxury vehicle segments as statistical outliers.
- **Business Intelligence Extraction:** Converted raw numerical analysis into understandable business-level insights.
- **Multivariate Analysis Capability:** Combined multiple categorical and numerical variables for deeper interpretation.
- **Strong Use of Python Libraries:** Demonstrated practical implementation of pandas, matplotlib, and seaborn for real-world analytics.
- **Data Interpretation Skills:** Generated meaningful interpretations instead of only visual outputs.
- **Professional Reporting:** Structured the analysis in a way suitable for academic projects, portfolios, and internship demonstrations.

Tools & Technologies Used

Technology	Purpose
Python	Programming Language
Pandas	Data Cleaning & Manipulation
Matplotlib	Visualization
Seaborn	Statistical Visualization
Jupyter Notebook	Development Environment

Project Conclusion

This project successfully demonstrated the practical application of Exploratory Data Analysis in understanding automobile market trends and pricing behavior. Through structured preprocessing, visual analysis, and statistical interpretation, the project transformed raw data into meaningful insights. The analysis highlighted the importance of

engine size, horsepower, body style, and drive-wheel configuration in determining vehicle prices. The project also demonstrated strong analytical thinking, data storytelling ability, and real-world data handling skills.

Outcome: This project can be effectively used in data analytics portfolios, internship applications, placement interviews, and academic submissions to demonstrate practical EDA skills.